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Wired For *Oops*

Neural Correlates of Loss of Control from a Car Racing Game Paradigm

Johns Hopkins University

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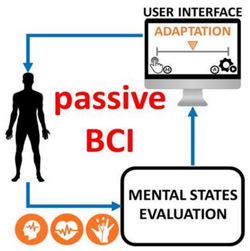
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Why & What?

Brain-computer interfaces (BCIs) represent a revolutionary bridge between the human brain and external devices, allowing for direct communication without the need for peripheral neural pathways.

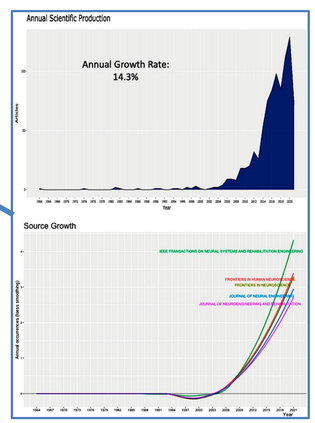


Noninvasive BCIs for communication



Among the varied BCI paradigms, passive BCIs (pBCIs) stand out as they do not require the user to execute a specific command but instead focus on interpreting the user's natural cognitive state.

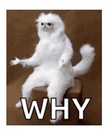
Rising Interest in EEG for Neurorehabilitation



Understanding the neural mechanisms of motor errors through pBCIs can offer a novel perspective for neurorehabilitation and next-generation BCIs.

Objective

This study aimed to elucidate the neural correlates associated with motor errors by observing **β-ERS** (Event-Related Synchronization) dynamics during unexpected machine behaviors using a pBCI car racing game.

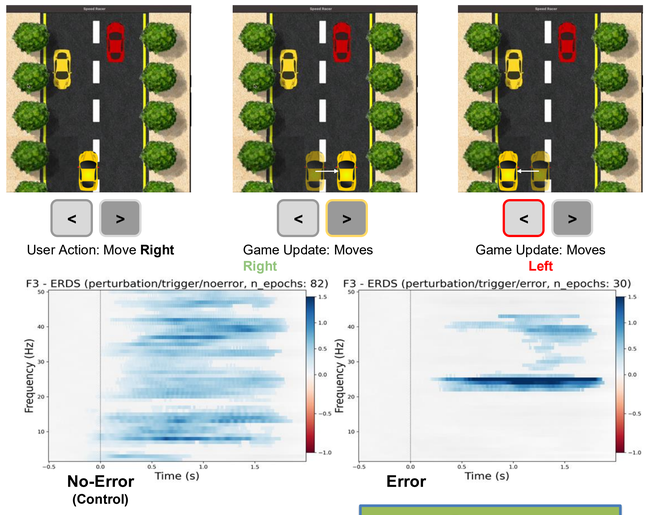


β??

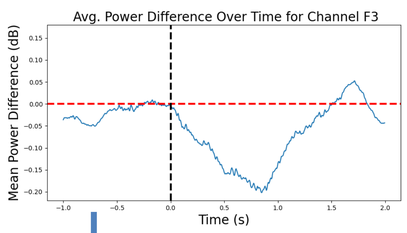
But… what did we actually do?

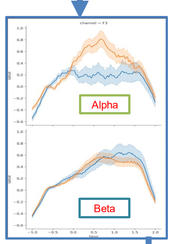
Paradigm Design

Participants played a simplified car racing game wherein a designed perturbation state caused the car to move in the opposite direction of the user's input (i.e., pressing the left key resulted in the car moving right, and vice-versa).
This unexpected behavior was introduced to simulate the loss of volitional control over the game. The neural responses, specifically β-ERS, were then recorded and analyzed to investigate any discernible patterns associated with these error trials contrasted to control trials where no such errors were introduced.

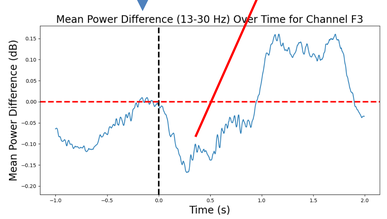


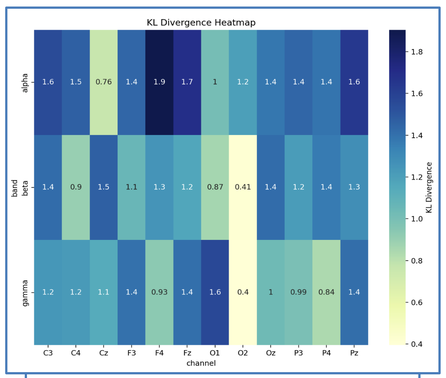
Results





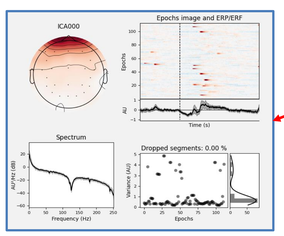
notable reduction in β-ERS during trials where motor errors were present



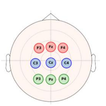


KL-Divergence Plot

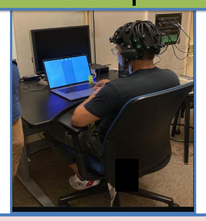
ERF data was normalized to ensure that each unique combination of channel and band for each condition represented a valid probability distribution.



Our main analysis focused on selected channels over the frontal (F3, Fz, F4), motor (C3, Cz, C4) and parietal regions (P3, Pz P4)



Setup



Analysis Pipeline

- Single participant with 4 control and 5 error sessions (~3 mins each)
- First and last 15s of each run clipped
- Approx. 10 mins of usable EEG data

1. Signal Processing
   1. Notch out 60, 120, 240 Hz
   2. High pass filter (0.1 – Hz)
2. Data Compilation
   - [tmin=-1, tmax=2] seconds
   - Removed all non-error events within 200ms of an error event
3. Drop bad channels: AF7, AF8, T7
4. Advanced Processing
   - FastICA on time-locked events
   - Drop IC for eye blinks
   - Project back to channel domain

What does it mean though?

Discussion



Our exploration into the dynamics related to error events has revealed a significant use for passive Brain-Computer Interfaces (pBCIs):

- Initial findings indicate a link between a participant's recognition of losing control, triggered by unanticipated actions of the machine and a reduced β-ERS response.
- This highlights the potential of employing pBCI to identify user states associated with the feeling of system error.
- Leveraging these insights can improve the communication and collaboration between the user and the machine.

Next Steps/Future Work

- Ensure game stability and add more realistic game scenarios
- Increase participant pool for data collection, targeting ~20 additional subjects.
- Develop and train a model for predicting error detection.
- Gather data from prosthetic users for integration into real-time prosthetic error correction systems.



Movement Decoding



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